Comparative Analysis of Monotonically Enforced Univariate Binning Strategies in Classification Using Logistic Regression

by: Zakaria Sherif, Dr. Herman Ray, Dr. Jonathan Boardman, Dr. Joseph White, Dr. Linh Le

Kennesaw State University

Research Overview

What is Feature Binning?

Feature Binning is technique that converts continuous variables into categorical bins to improve interpretability and performance.

10 Methods tested

Dataset: KSU Equifax dataset

Key Metrics Evaluated

- Kolmogorov-Smirnov (KS Statistic): distribution separation
- GINI Coefficient (GINI): discrimination ability.
- Average Information Value (IV): Predictive power measure

Research Methodology: Pipeline Overview

Preprocessing

Data cleaning, and initial variable reduction

Binning Transformation

Apply various binning methods, enforce monotonicity, calculate Weight of Evidence

Feature Selection

VIF selection and AIC-based backward elimination in two phases

Model Training

Train logistic regression model on transformed data

Evaluation

Calculate GINI, KS Statistic, and IV

Binning Transformation Process

Pipeline Creation

For each feature, create a pipeline with Binning Transformation, Monotonicity Enforcer, and WOE Transformation

Information Value Calculation

Calculate IV for each feature using the fitted WOE Transformation

Feature Transformation

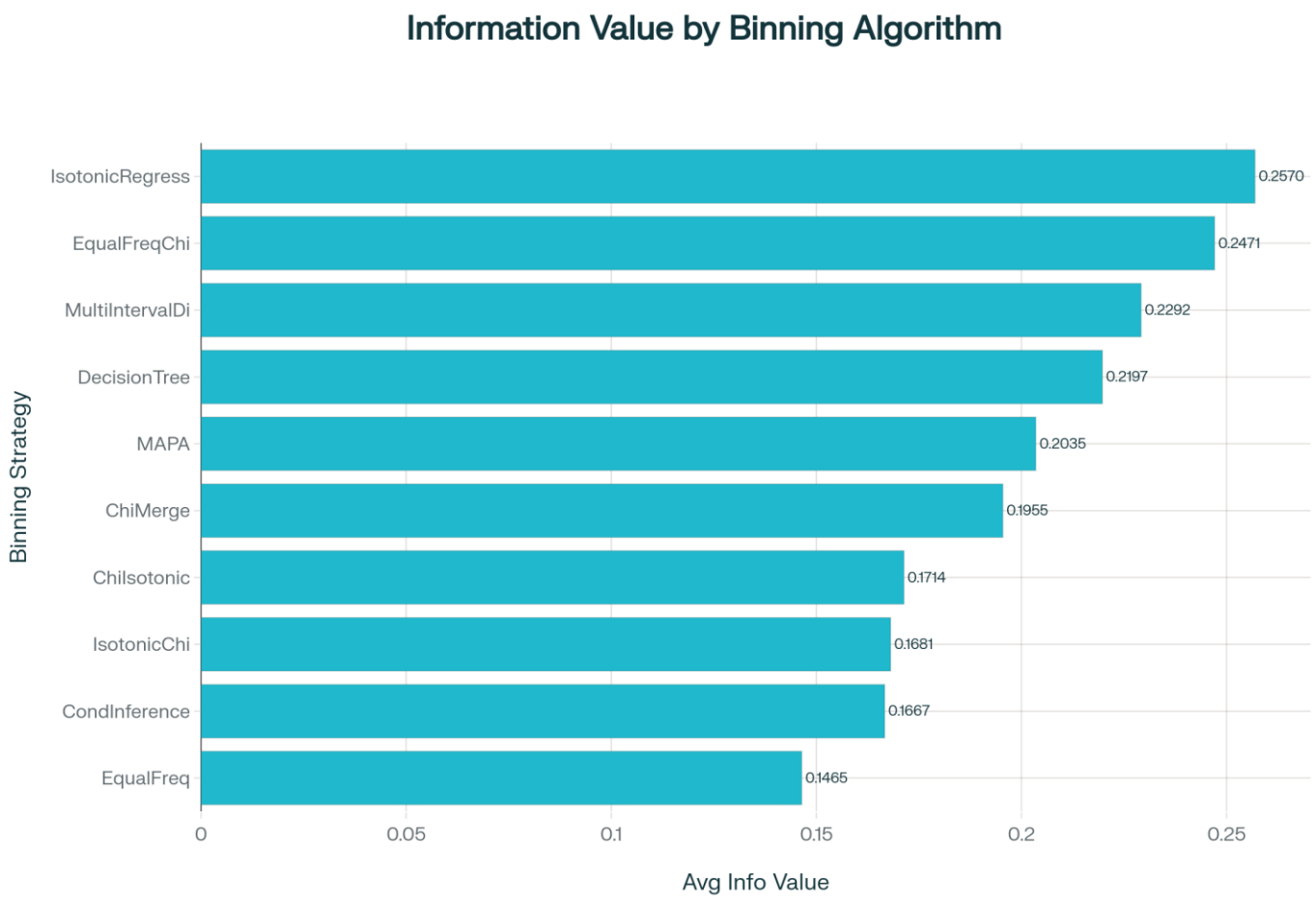
Transform features applying binning and monotonicity steps

Combination & Selection

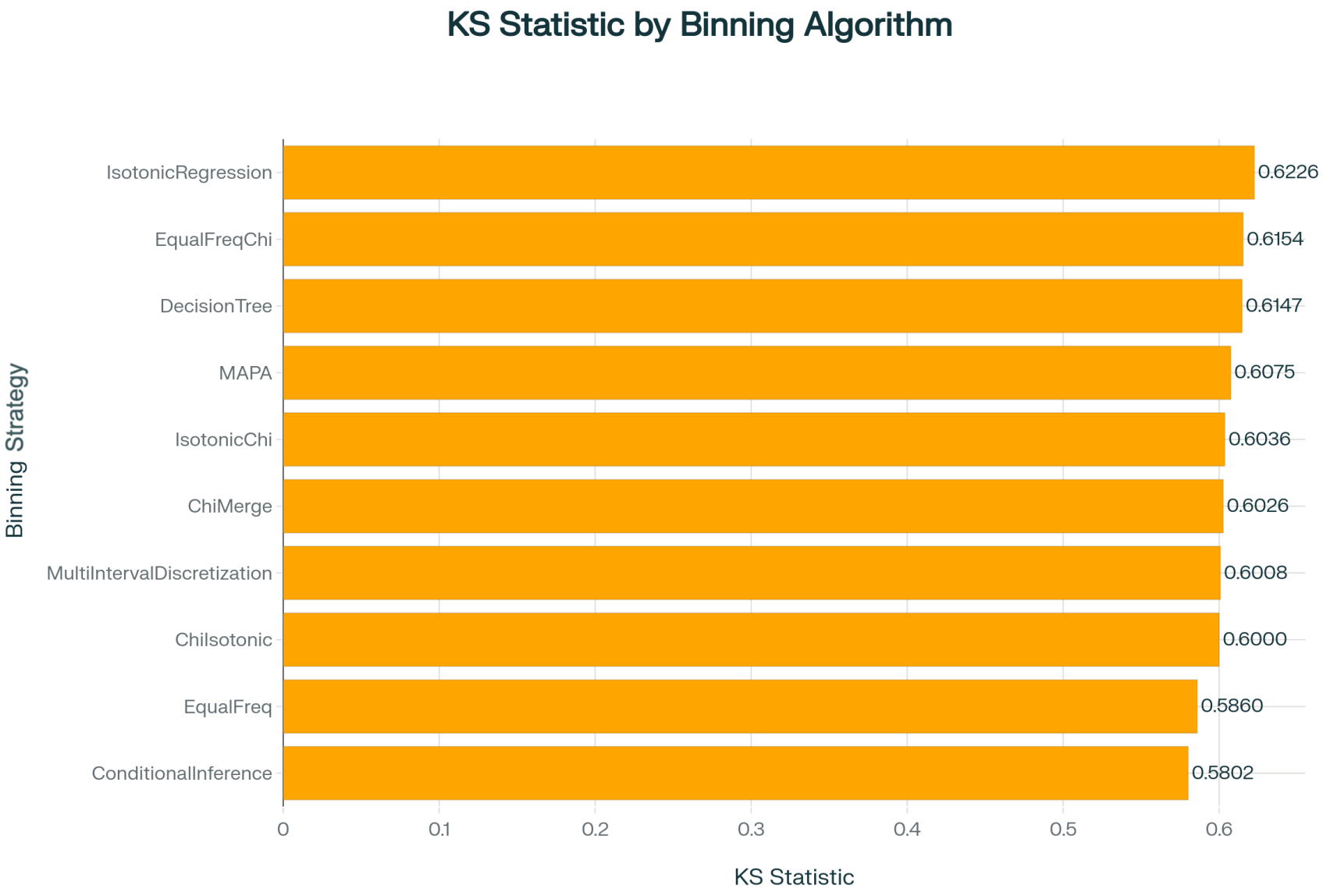
Combine transformed features and select the most predictive ones using VIF and Stepwise elimination

Preliminary Results

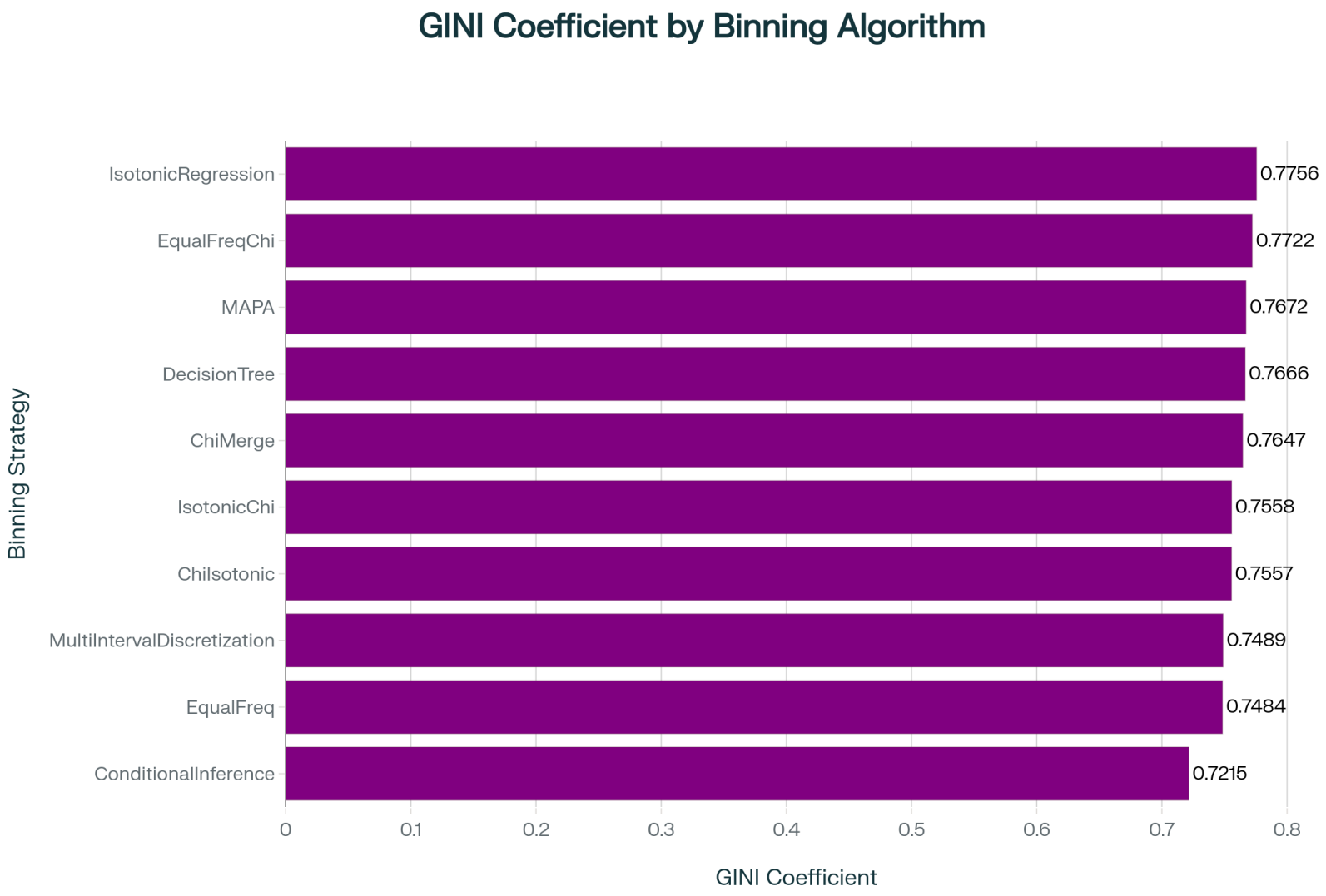
Information Value Results



KS Results


KS Statistic by Binning Algorithm
Binning Strategy
IsotonicRegression
EqualFreqChi
DecisionTree
MAPA
IsotonicChi
ChiMerge
MultiIntervalDiscretization
ChiIsotonic
EqualFreq
ConditionalInference
0.6226
0.6154
0.6147
0.6075
0.6036
0.6026
0.6008
0.6000
0.5860
0.5802
0
0.1
0.2
0.3
0.4
0.5
0.6
KS Statistic

GINI Results



Key Findings

- Overall Best Performer

  Isotonic Regression achieved perfect #1 composite ranking, leading in 3 out of 3 key metrics with exceptional consistency

- Consistently Strong

  Equal Frequency-Chi ranked #2 overall with strong performance across all metrics and reasonable computation efficiency

- High Correlation

  KS and GINI show high correlation (r=0.914), confirming strong relationship between discrimination metrics

Conclusion

- All algorithms show good discrimination (GINI > 0.7215), providing flexibility in algorithm choice.

- Isotonic Regression has overall best predictive performance, leading in 3 out of 3 key metrics

- Equal Frequency-Chi Square is a very close second and an excellent alternative with consistent performance across all metrics; it has the advantage of efficiency, simplicity and statistical power.

What is next?

- Finalizing the Results
- Publishing at NCUR National Conference on Undergraduate Research

Questions?